

Programming Assignment #1: Slope Intercept Form of a Line

1 Overview

Write a simple C program to calculate the slope-intercept form of a line equation. The program should prompt the user for the coordinates of the first point, x first, and then y , and then prompt the user for the second point, x first and then y . After that, the program should calculate and then print out the slope and y -intercept of the line and equation of the line in slope-intercept form $y = mx + b$.

The program should then prompt the user for the x coordinate of a third point and then print out the corresponding y coordinate for that point.

Your program should

- Request the user input two coordinates one number at a time. (For the user to input (x_1, y_1) and (x_2, y_2) , the program needs to request each number separately.)
- Calculate the slope of the line and the y -intercept of the line.
- Output the equation in the form $y = mx + b$ substituting the values you calculated for the slope and y -intercept.
- Request the user input a third x value.
- Calculate the y value for this input.
- Print the solution to the screen.

2 Code Documentation

The very first lines in *all* of your programming project files this semester should be comments in the class header file (copy from header.c). Of course, the items in the square brackets `[]` in the comments should be filled in with the proper information.

In addition, the first output for all of your projects this semester, both for the lab exercises and the programming assignments, should be the name of the project and your name, e.g:

```
printf("Programming Assignment #1 - Joe Student\n");
```

Your code should meet all the stylistic requirements we have discussed in class including the appropriate use of whitespace and comments.

3 Sample Run

Your output should match the sample run below.

```
Mark Schwitzerlett -- Programming Assignment #1

Please input the first coordinate
The x value is -4.75
The y value is 2
Please input the second coordinate
The x value is 6
The y value is -3.5

The first point is (-5, 2) and the second point is (6, -4)
The slope is -0.51
The y-intercept is -0.43
The slope intercept equation of the line is  $y = -0.51x + -0.43$ 

Please input the x coordinate of a third point 8

The corresponding y value is -4.52

...Program finished with exit code 0
Press ENTER to exit console.█
```

4 Deliverables

You should turn in a stand-alone, complete application program (your C source code, a single “.c” file) containing a `main()` function. All labs and projects this semester will be submitted in Gradescope (via a Canvas assignment).

Due date: The assignment is due by 11:59 pm Sunday, September 14th.